

SIMPLE PROOF OF A GENERAL MATRIX EQUALITY

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Summary: In this paper a simple and self-contained proof of a general matrix equality is presented. It was introduced by Shapiro (1987).

1. INTRODUCTION

The purpose of this note is to present a simple and self-contained proof of a general matrix equality which, to our knowledge, was introduced by Shapiro (1987, p. 59). The equality also plays a fundamental role in Browne and Shapiro (1988; see equation (A.8)), and a restricted version is used in Satorra and Bentler (1990, equality (A.5)).

Shapiro (1987) uses a matrix equality of Rao (1973, p. 33) to prove a restricted version of the equality for non-singular matrices Ω and C defined below, and the result is extended to singular matrices "by the argument of continuity" (p. 59).

In contrast to Shapiro's work, our proof is direct, does not invoke previous matrix equalities, and does not require the somewhat loose "argument of continuity".

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2. THE MATRIX EQUALITY AND PROOF

For conformable matrices Δ , Ω and C , we have

$$(1) \quad (\Delta'(\Omega + \Delta C \Delta')^{-1} \Delta)^+ = (\Delta' \Omega^{-1} \Delta)^+ + C$$

provided that

- (i) Ω and C are positive semidefinite (and hence symmetric);
- (ii) $M(\Omega) \supset M(\Delta)$, $M(\Omega + \Delta C \Delta') \supset M(\Omega)$; and
- (iii) $M(\Delta') \supset M(C)$.

Here M denotes column space, $(\cdot)^{-}$ denotes any g-inverse (i.e., $AA^{-}A = A$ for matrix A) and $(\cdot)^{+}$ denotes the Moore-Penrose generalized inverse.

Proof

Essential for proving (1) is the following equality:

$$(2) \quad \Omega(\Omega + H)^{-}(I + H\Omega^{-})\Omega = \Omega,$$

where $H = \Delta C \Delta'$.

The left-hand side of the equality is equal to

$$\begin{aligned} & \Omega(\Omega + H)^{-}\Omega + \Omega(\Omega + H)^{-}H\Omega^{-}\Omega = \\ & \Omega(\Omega + H)^{-}\Omega + \Omega(\Omega + H)^{-}H = \Omega(\Omega + H)^{-}(\Omega + H) = \Omega, \end{aligned}$$

since $M(\Omega + H) \supset M(\Omega) \supset M(\Delta) \supset M(H)$, and Ω and $\Omega + H$ are symmetric matrices.

Let $\Delta = \Omega A$. Then

$$\begin{aligned} (3) \quad & \Delta' \Omega^{-} \Delta - \Delta' (\Omega + H)^{-} \Delta - \Delta' (\Omega + H)^{-} H \Omega^{-} \Delta = \\ & A' (\Omega \Omega^{-} \Omega - \Omega (\Omega + H)^{-} \Omega - \Omega (\Omega + H)^{-} H \Omega^{-} \Omega) A = \\ & A' (\Omega - \Omega (\Omega + H)^{-} (I + H \Omega^{-}) \Omega) A = 0. \end{aligned}$$

Finally (1) follows from

$$(\Delta' (\Omega + H)^{-} \Delta)^+ (\Delta' \Omega^{-} \Delta - \Delta' (\Omega + H)^{-} \Delta - \Delta' (\Omega + H)^{-} H \Omega^{-} \Delta) (\Delta' \Omega^{-} \Delta)^+ = 0,$$

because (i), (ii) and (iii) imply that $M(\Delta' (\Omega + H)^{-} \Delta) = M(\Delta' \Omega^{-} \Delta) \supset M(C)$.

We further used the property that $M(B^+) = M(B)$ for symmetric B .

3: REMARKS

- Note that conditions (ii) of the lemma are trivially fulfilled when Ω and $\Omega + \Delta C \Delta'$ are nonsingular. Condition (iii) is fulfilled when $\Delta' \Omega^{-1} \Delta$ is nonsingular.
- Setting $C := JQJ'$, where $J := [0, I, 0]'$ and 0 and I denote zero and identity matrix, respectively, of conformable dimensions (the numbers of columns of the zero matrices may differ) and under the conditions of the lemma, with symmetric matrix Q , we have

$$(4) \quad (\Delta(\Omega + \Delta_2 Q \Delta_2')^{-1} \Delta)^+ = (\Delta' \Omega^{-1} \Delta)^+ + JQJ',$$

where $\Delta_2 := \Delta J$.

- From $M(\Omega) \supset M(\Delta)$ and expansion and collection of terms on the left-hand side it is simple to verify

$$(5) \quad (\Delta' \Omega^{-1} \Delta)^+ \Delta' \Omega^{-1} (\Omega + \Delta C \Delta') \Omega^{-1} \Delta (\Delta' \Omega^{-1} \Delta)^+ = (\Delta' \Omega^{-1} \Delta)^+ + C.$$

- We further have

$$(6) \quad (\Delta' (\Omega + \Delta C \Delta')^{-1} \Delta)^{-1} = (\Delta' \Omega^{-1} \Delta)^{-1} + C,$$

provided that

- (i*) Ω and C are symmetric;
- (ii) $M(\Omega) \supset M(\Delta)$, $M(\Omega + \Delta C \Delta') \supset M(\Omega)$; and
- (iii*) the two inverses exist.

The proof hinges on equality (2) and its corollary (3).

Clearly (1) is more general than (6), but for this a price has to be paid. The price is that Ω and C have to be positive semi-definite, although there is some compensation as condition (iii) is weaker than (iii*).

The equalities (1), (4), (5) and (6) simplify the asymptotic variance matrix of estimates in covariance-structure analysis. See in this respect the references cited above and also Shapiro (1986) for the case of a singular information matrix.

Note

Comments of an anonymous referee are gratefully acknowledged. They have led to a considerably more general result.

4. REFERENCES

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